

THE GLAMSQUAD SALON

Data Model & Dax Measures

Power BI Portfolio Rebuild | Profitability & Operational Analytics

Document Control	Details
Client	The GlamSquad Salon
Role	Business Analyst + Data Analyst
Project Period	May–Aug 2025
Confirmed Tool	Power BI dashboard built from scratch
Model Status	Recommended logical portfolio rebuild
DAX Status	Reconstructed formulas for portfolio demonstration; not claimed as historical DAX
Purpose	Define a defensible star-style analytical model and a reusable DAX measure library aligned to the KPI Catalogue and Power BI Requirements.
Critical evidence boundary: The exact historical PBIX model, table names, relationships, Power Query steps and DAX formulas are not retained. Everything in this document that specifies exact schema or DAX is a portfolio reconstruction designed to reproduce the confirmed analytical intent without fabricating history.	

1. Modelling Objective

The portfolio model should support profitability, service/pricing, staff/utilization, customer/repeat, appointment and feedback analysis with consistent filter behaviour. A star-style model is recommended because it makes business measures easier to validate and reduces ambiguous relationships.

Model design principle: Dimensions describe who/what/when; fact tables record business events or measurable transactions.

2. Recommended Logical Model

Entity	Type	Recommended Grain	Primary Analytical Use
DimDate	Dimension	One row per calendar date	Time filtering and trends
DimCustomer	Dimension	One sanitized row per logical customer	Repeat customer and retention analysis
DimService	Dimension	One row per standardized service	Service mix, pricing and service economics
DimStaff	Dimension	One sanitized row per stylist/staff member	Stylist performance and cost context
FactAppointments	Fact	One row per appointment/booking record	Appointments, revenue, discounts, cancellations/no-shows
FactStaffHours	Fact	One row per staff/date or working period	Working hours and utilization context
FactExpenses	Fact	One row per expense transaction	Expenses and profit calculation
FactFeedback	Fact-like qualitative	One sanitized row per feedback item	Theme/source analysis
Naming note: These exact table names are proposed portfolio names, not retained historical names.			

3. Recommended Relationship Pattern

From	To	Cardinality	Filter Direction	Reason
DimDate[Date]	FactAppointments[AppointmentDate]	1:*	Single	Appointment/revenue time analysis
DimCustomer[CustomerKey]	FactAppointments[CustomerKey]	1:*	Single	Customer visit/repeat analysis
DimService[ServiceKey]	FactAppointments[ServiceKey]	1:*	Single	Service analysis
DimStaff[StaffKey]	FactAppointments[StaffKey]	1:*	Single	Stylist attribution
DimDate[Date]	FactStaffHours[WorkDate]	1:*	Single	Working-hours analysis
DimStaff[StaffKey]	FactStaffHours[StaffKey]	1:*	Single	Staff capacity analysis
DimDate[Date]	FactExpenses[ExpenseDate]	1:*	Single	Expense time analysis
Relationship control: Avoid unnecessary bidirectional relationships. Optional cost attribution to service/staff should only be modelled when the underlying sanitized data provides a defensible key.				

4. Core Table Design

4.1 FactAppointments

Field	Type	Role
AppointmentID	Key	Unique logical appointment identifier
AppointmentDate	Date/FK	Connects to DimDate
CustomerKey	FK	Sanitized customer relationship
ServiceKey	FK	Service relationship
StaffKey	FK	Stylist/staff relationship
AppointmentStatus	Text	Completed / Cancelled / No-show / unresolved standardized status
RevenueAmount	Decimal	Validated realized revenue where supported
DiscountAmount	Decimal	Validated discount amount where supported
ListedValue	Decimal	Optional portfolio field where source supports listed value
DurationMinutes	Whole Number	Optional, only if available in rebuild data

4.2 Dimensions / Supporting Facts

Table	Important Fields	Notes
DimCustomer	CustomerKey, PhoneMatchKey (protected), FirstVisitDate, LastVisitDate	Never expose raw phone numbers publicly.
DimService	ServiceKey, ServiceName, ListedPrice, DirectMaterialCost (conditional)	Exact prices/costs must be sanitized.
DimStaff	StaffKey, StaffDisplayName, SalaryCost, Commission context	Use aliases/sanitized labels publicly.
FactStaffHours	StaffKey, WorkDate, ActualWorkingHours, ScheduledHours (optional), BookedServiceTime (optional)	Utilization only if denominator is defensible.
FactExpenses	ExpenseID, ExpenseDate, ExpenseCategory, ExpenseAmount	Avoid double counting salary/commission/material costs if already embedded elsewhere.
FactFeedback	FeedbackID, FeedbackDate, FeedbackSource, FeedbackTheme	Qualitative table remains separate from financial facts.

5. Date Table

A dedicated date dimension is recommended for stable time intelligence. The exact historical calendar implementation is unknown.

```
DimDate =
ADDCOLUMNS (
    CALENDAR ( MIN ( FactAppointments[AppointmentDate] ), MAX ( FactAppointments[AppointmentDate] ) ),
    "Year", YEAR ( [Date] ),
    "Month Number", MONTH ( [Date] ),
    "Month Name", FORMAT ( [Date], "MMM" ),
    "Year Month", FORMAT ( [Date], "YYYY-MM" ),
    "Day Name", FORMAT ( [Date], "DDD" ),
    "Day of Week Number", WEEKDAY ( [Date], 2 )
)
```

Portfolio note: The date table is proposed DAX for the rebuild. It is not historical client code.

6. DAX Measure Design Rules

- Prefer measures over calculated columns for filter-sensitive aggregations.
- Use DIVIDE() instead of / for percentage measures to handle zero denominators.
- Base every KPI on cleaned/standardized status and key logic.
- Keep business definitions in one measure rather than recreating formulas inside visuals.
- Do not calculate utilization, margin or retention when the required denominator/eligibility rule is not defensible.
- Use sanitized portfolio values and identifiers only in public outputs.

7. Base Measures

```
Total Revenue =
SUM ( FactAppointments[RevenueAmount] )

Total Expenses =
SUM ( FactExpenses[ExpenseAmount] )

Profit =
[Total Revenue] - [Total Expenses]

Profit Margin % =
DIVIDE ( [Profit], [Total Revenue] )
```

Important modelling decision: If salary, commission or material costs are stored both in FactExpenses and separate staff/service tables, establish one authoritative expense treatment before calculating Total Expenses to prevent double counting.

8. Profit Improvement Measure

The engagement outcome retained for the case study is a 15% profit increase within approximately two months. Underlying rupee values are confidential. A portfolio before/after implementation requires an explicit period design.

```
Profit Improvement % =
VAR ProfitBefore = [Profit Before]
VAR ProfitAfter  = [Profit After]
RETURN
    DIVIDE ( ProfitAfter - ProfitBefore, ProfitBefore )
```

Do not fabricate: [Profit Before] and [Profit After] require documented period filters or a scenario table in the sanitized rebuild. Do not invent historical implementation dates or hard-code the 15% result into DAX.

9. Appointment Measures

```
Total Appointments =
COUNTROWS ( FactAppointments )

Completed Appointments =
CALCULATE (
    [Total Appointments],
    FactAppointments[AppointmentStatus] = "Completed"
)

Cancelled Appointments =
CALCULATE (
    [Total Appointments],
    FactAppointments[AppointmentStatus] = "Cancelled"
)

No-Show Appointments =
CALCULATE (
    [Total Appointments],
    FactAppointments[AppointmentStatus] = "No-show"
)
```

Status labels: The exact source status labels are not retained. "Completed", "Cancelled" and "No-show" are proposed standardized portfolio categories.

10. Cancellation & No-Show Rates

```
Eligible Booked Appointments =
[Completed Appointments] + [Cancelled Appointments] + [No-Show Appointments]

Cancellation Rate % =
DIVIDE ( [Cancelled Appointments], [Eligible Booked Appointments] )

No-Show Rate % =
DIVIDE ( [No-Show Appointments], [Eligible Booked Appointments] )
```

Definition boundary: The exact historical denominator is not retained. This is a documented portfolio rule and should be presented as such.

11. Customer & Repeat Measures

```
Unique Customers =
DISTINCTCOUNT ( FactAppointments[CustomerKey] )

Repeat Customers =
COUNTROWS (
    FILTER (
        VALUES ( FactAppointments[CustomerKey] ),
        CALCULATE ( [Completed Appointments] ) > 1
    )
)

Repeat Customer Rate % =
DIVIDE ( [Repeat Customers], [Unique Customers] )
```

Confirmed logic: Phone-number data was used to identify repeat visits. In the portfolio, raw phone values should be transformed into sanitized customer keys before Power BI.

12. Retention Measure — Requires Rebuild Rule

The project included customer-retention analysis, but the exact historical retention window/eligibility definition is not retained. Therefore, a final DAX measure should only be implemented after the sanitized portfolio defines the business rule.

```
Customer Retention Rate % =
-- PSEUDOCODE / REBUILD PLACEHOLDER
-- Retained eligible customers / eligible baseline customers
-- Exact time window and eligibility rule must be documented first.
```

Interview answer: Explain the historical analytical objective confidently, but say the exact retention window is not retained. For the public rebuild, you define and document a transparent rule rather than pretending to remember it.

13. Service Measures

```
Revenue by Service =
[Total Revenue]
-- Place DimService[ServiceName] on the visual axis

Service Count =
[Completed Appointments]

Service Mix % =
DIVIDE (
    [Completed Appointments],
    CALCULATE ( [Completed Appointments], REMOVEFILTERS ( DimService ) )
)

Average Revenue per Completed Appointment =
DIVIDE ( [Total Revenue], [Completed Appointments] )
```

14. Discount Measures

```
Total Discount Amount =
SUM ( FactAppointments[DiscountAmount] )
```

```
Discount Rate % =
DIVIDE ( [Total Discount Amount], [Potential / Listed Value] )
```

Conditional denominator: Discount Rate % should only be used if the sanitized data contains a reliable listed/pre-discount value. Do not infer this denominator from revenue alone.

15. Service Contribution / Margin

Service economics were part of the engagement. Exact historical cost-allocation logic is not retained.

```
Direct Service Cost =
SUM ( FactAppointments[DirectMaterialCost] )
+ SUM ( FactAppointments[DirectCommissionCost] )
```

```
Service Contribution =
[Total Revenue] - [Direct Service Cost]
```

```
Service Margin % =
DIVIDE ( [Service Contribution], [Total Revenue] )
```

Use only when defensible: The exact fact-table placement of direct cost fields is illustrative. If reliable direct costs are unavailable, suppress the margin measure rather than allocate costs arbitrarily.

16. Staff Measures

```
Revenue by Stylist =
[Total Revenue]
-- Place DimStaff[StaffDisplayName] on the visual axis
```

```
Appointments by Stylist =
[Completed Appointments]
```

```
Total Working Hours =
SUM ( FactStaffHours[ActualWorkingHours] )
```

```
Revenue per Working Hour =
DIVIDE ( [Total Revenue], [Total Working Hours] )
```

Interpretation rule: Do not use Revenue per Working Hour by itself to label a stylist as high or low performance. Combine with appointments, working hours, service mix, repeat behaviour and qualitative context where available.

17. Staff Utilization

```
Booked Service Hours =
DIVIDE ( SUM ( FactStaffHours[BookedServiceMinutes] ), 60 )
```

```
Available Staff Hours =
SUM ( FactStaffHours[AvailableWorkingHours] )
```

```
Staff Utilization % =
DIVIDE ( [Booked Service Hours], [Available Staff Hours] )
```

Critical condition: This measure is valid only if the portfolio has reliable booked-service time and available-capacity hours. Historical utilization was analyzed, but the exact original denominator is not retained.

18. Commission & Material Cost Measures

```
Commission Cost =
SUM ( FactAppointments[CommissionAmount] )
```

```
Commission to Revenue % =
DIVIDE ( [Commission Cost], [Total Revenue] )
```

```
Material Cost =
SUM ( FactAppointments[MaterialCost] )
```

```
Material Cost % =
DIVIDE ( [Material Cost], [Total Revenue] )
```

Schema note: The fields above may instead live in FactExpenses or a separate cost table. The exact historical structure is unknown. Choose one authoritative design in the portfolio and avoid double counting.

19. Feedback Measures

```
Feedback Items =
COUNTROWS ( FactFeedback )
```

```
Feedback Theme Frequency =
COUNTROWS ( FactFeedback )
-- Place FactFeedback[FeedbackTheme] on the visual axis
```

Evidence boundary: No historical feedback counts or sentiment percentages are retained. In a sanitized rebuild, this measure is appropriate only for the feedback records actually included.

20. Demand / Peak-Period Analysis

The retained qualitative pattern is stronger demand generally in evenings and around major-festival periods, but exact historical peak hours/days are not retained. The portfolio should let the sanitized data determine the pattern.

```
Appointments by Hour =
[Total Appointments]
-- Use a rebuilt TimeBand/Hour field on the visual axis
```

```
Appointments by Day =
[Total Appointments]
-- Use DimDate[Day Name] with correct sort order
```

No hard-coded peak: Do not create a measure that labels a specific historical hour/day as peak unless the sanitized rebuild data actually demonstrates it.

21. Suggested Measure Table

Measure Group	Measures
Profitability	Total Revenue, Total Expenses, Profit, Profit Margin %, Profit Improvement %
Appointments	Total, Completed, Cancelled, No-Show, Eligible Booked, Cancellation Rate %, No-Show Rate %
Customers	Unique Customers, Repeat Customers, Repeat Customer Rate %, Retention Rate % (after rule definition)
Services	Revenue by Service, Service Count, Service Mix %, Avg Revenue/Completed Appointment, Discount measures, Service Contribution/Margin
Staff	Revenue by Stylist, Appointments by Stylist, Total Working Hours, Revenue/Working Hour, Staff Utilization %, Commission Cost/Ratio
Costs	Material Cost, Material Cost %
Feedback	Feedback Items, Feedback Theme Frequency

22. Measure Naming Standards

- Use business-readable names rather than technical abbreviations.
- Append % only to ratio/percentage measures.
- Keep raw columns hidden from report users when they are not needed.
- Group measures in display folders such as Profitability, Appointments, Customers, Services, Staff and Costs.
- Document any conditional measure whose validity depends on source completeness.

23. DAX Validation Tests

Test	Expected Result
Profit arithmetic	[Profit] equals [Total Revenue] - [Total Expenses] under identical filters.
Cross-page consistency	The same measure returns the same result under equivalent filter context.
Zero denominator	Rate measures return blank/controlled result rather than error/infinity.
Status logic	Completed/Cancelled/No-show measures match cleaned status counts.
Repeat customer	Sample customers with one vs multiple completed visits classify correctly.
Service mix	Service Mix % sums to the intended total under the documented scope.
Staff revenue	Stylist totals reconcile to attributed appointment revenue.
Utilization	Measure remains unavailable if valid capacity denominator is missing.
Cost measures	Salary/commission/material costs are not double counted.
Privacy	No PII or confidential original values are exposed.

24. Common Modelling Risks

Risk	Why It Matters	Control
Double-counted costs	Inflates expenses and understates profit.	Define one source of truth for each cost category.
Many-to-many joins	Can create duplicated values.	Use dimensions/bridge only when analytically justified.
Bidirectional filters	Can produce ambiguous results.	Prefer single-direction relationships.
Unclean status labels	Distorts cancellation/no-show KPIs.	Standardize before load.
PII in model	Creates confidentiality risk.	Use sanitized keys; exclude phone numbers from public PBIX.
Hard-coded 15%	Would fake the result rather than calculate it.	Calculate from before/after portfolio periods if present.
Unsupported utilization	Produces misleading efficiency percentages.	Suppress until denominator is defensible.

25. Mapping to Dashboard Pages

Dashboard Page	Primary Measures
Executive Profitability Overview	Total Revenue, Total Expenses, Profit, Profit Margin %, Profit Improvement %
Service & Pricing Analysis	Revenue by Service, Service Count, Service Mix %, Avg Revenue/Appointment, Discount measures, Service Contribution/Margin
Staff Performance & Utilization	Revenue by Stylist, Appointments by Stylist, Working Hours, Revenue/Working Hour, Staff Utilization %, Commission measures
Customer & Appointment Analysis	Appointments, Cancellation/No-Show Rates, Unique/Repeat Customers, Repeat Rate, Retention (defined rebuild rule), demand analysis

26. Confirmed vs Reconstructed

Classification	Details
Confirmed	Power BI dashboard built from scratch; analyses included profitability, appointment utilization, service mix, repeat rate/retention, peak demand, staff performance, revenue by service/stylist, cancellations/no-shows, pricing, staff/commission/material costs and feedback.
Confirmed logic	Profit = Revenue - Expenses; phone-number data used for repeat-customer matching; 15% profit improvement within approximately two months.
Reconstructed portfolio design	Exact table names, surrogate keys, relationships, date table, measure names and DAX formulas in this document.
Not retained / not claimed	Exact historical PBIX schema, original DAX, Power Query M, precise utilization/retention denominator, exact cost-allocation method, refresh architecture or original dashboard page structure.